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Laminate, method of producing laminate, method of forming optical body, and camera module-equipped device

Abstract

A laminate includes a thin film structural body and retention films. A first retention film is stacked on one face of the thin film structural body and a second retention film is stacked on another face of the thin film structural body. The thin film structural body has a first fine irregularity structure at a face that is in contact with the first retention film and has a second fine irregularity structure at a face that is in contact with the second retention film. The first retention film has a third fine irregularity structure at a face that is in contact with the thin film structural body. The second retention film has a fourth fine irregularity structure at a face that is in contact with the thin film structural body. The laminate includes a half cut section. The second retention film contains a UV-curable resin.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a laminate, a method of producing a laminate, a method of forming an optical body, and a camera module-equipped device.

BACKGROUND

(2) In display devices such as liquid-crystal displays and optical devices such as cameras, an incident surface of light of a substrate such as a display panel or a lens is often subjected to anti-reflection treatment in order to avoid deterioration of visibility and image quality (occurrence of color irregularities, ghosts, etc.) caused by reflection of extraneous light. This anti-reflection treatment may be performed by a conventionally known method in which a fine irregularity structure is formed at the incident surface of light so as to reduce reflectance.

(3) Formation of a fine irregularity structure at the incident surface of light can be achieved by, for example, affixing a film having a fine irregularity structure at a surface thereof to the incident surface. This method is highly advantageous from viewpoints such as that processing of the substrate itself is not necessary, that the film itself can be distributed to market (i.e., is portable), and that a fine irregularity structure can be formed in a partial manner in a target region at the surface of the substrate.

(4) In a situation in which a film having anti-reflection function is itself to be distributed, performing half cut processing of a film in which peelable films are stacked makes it is easy to peel off a required part when the film having anti-reflection function is to be bonded to an object serving as an adherend.

(5) For example, Patent Literature (PTL) 1 discloses a pressure-sensitive adhesive film for half cut processing that includes a substrate layer and a pressure-sensitive adhesive layer. In another example, PTL 2 discloses half cutting a functional optical film that is stacked on a conductive metal layer. In the invention disclosed in PTL 2, the functional optical film is affixed to the conductive metal layer via an adhesive.

CITATION LIST

Patent Literature

(6) PTL 1: JP 2017-149948 A PTL 2: JP 2004-151633 A

SUMMARY

Technical Problem

(7) The film disclosed in PTL 1 is affixed through a pressure-sensitive adhesive layer, and the film

disclosed in PTL 2 is affixed through an adhesive. A film that is affixed via a pressure-sensitive adhesive layer, an adhesive, or the like faces issues such as environment testing durability being low and waviness readily occurring.

(8) The present disclosure is directed to solving the various problems in the conventional art set forth above and achieving the following objects. Specifically, one object of the present disclosure is to provide a laminate that enables simple peeling of a thin film structural body having a fine irregularity structure at a surface thereof when the thin film structural body is to be bonded to an object serving as an adherend, that has improved environment testing durability, and in which waviness is inhibited. Another object of the present disclosure is to provide a method of producing this laminate. Another object of the present disclosure is to provide a method of forming an optical body having excellent anti-reflection performance on a base plate using this laminate. Another object of the present disclosure is to provide a camera module-equipped device that can acquire a captured image in which color irregularities, ghosts, and the like are inhibited.

Solution to Problem

(9) The following are provided as a solution to the problems set forth above.

(10) <1> A laminate comprising a thin film structural body and retention films, wherein a first retention film is stacked on one face of the thin film structural body and a second retention film is stacked on another face of the thin film structural body, the thin film structural body has a first fine irregularity structure at a face that is in contact with the first retention film and has a second fine irregularity structure at a face that is in contact with the second retention film, the first retention film has a third fine irregularity structure at a face that is in contact with the thin film structural body, the second retention film has a fourth fine irregularity structure at a face that is in contact with the thin film structural body, the laminate includes a half cut section in a thickness direction, and the second retention film contains a UV-curable resin, and the UV-curable resin has a storage modulus at 25° C. of 1.1 GPa or less.

(11) <2> The laminate according to <1>, wherein a base substrate of the first retention film has a thickness of 100 μm or less.

(12) <3> The laminate according to <1> or <2>, wherein $P1 > P2$ is satisfied when peeling force at an interface of the first retention film and the thin film structural body in a 90° peeling test in accordance with JIS Z0237:2009 is taken to be $P1$ (N/25 mm) and peeling force at an interface of the second retention film and the thin film structural body in a 90° peeling test in accordance with JIS Z0237:2009 is taken to be $P2$ (N/25 mm).

(13) <4> The laminate according to any one of <1> to <3>, wherein the half cut section has a circular shape or a polygonal shape in a top view of the laminate.

(14) <5> The laminate according to any one of <1> to <4>, wherein the thin film structural body has a thickness of 5 μm or less.

(15) <6> The laminate according to any one of <1> to <5>, wherein the first retention film contains a UV-curable resin.

(16) <7> The laminate according to any one of <1> to <6>, wherein the third fine irregularity structure is an inverted structure of the first fine irregularity structure and the fourth fine irregularity structure is an inverted structure of the second fine irregularity structure.

(17) <8> The laminate according to any one of <1> to <7>, wherein the first fine irregularity structure, the second fine irregularity structure, the third fine irregularity structure, and the fourth fine irregularity structure are each formed of an irregularity pattern having a pitch equal to or less than a wavelength of visible light.

(18) <9> A method of producing a laminate comprising a step of, in formation of the half cut section in the laminate according to any one of <1> to <8>, forming the half cut section with the laminate in a pressed state.

(19) <10> A method of forming an optical body on a base plate comprising peeling the first retention film and the second retention film from the laminate according to any one of <1> to <8>

and stacking the thin film structural body on a base plate via an adhesive.

(20) <11> The method of forming an optical body on a base plate according to <10> comprising: a first peeling step of peeling a single-side peeled laminate formed of the thin film structural body and the first retention film from the second retention film, inside of the half cut section, to obtain the single-side peeled laminate; an application step of applying the adhesive onto the base plate; a pressing step of pressing the single-side peeled laminate against the adhesive that has been applied such that a face where the second retention film has been peeled off faces toward the base plate; a curing step of irradiating the adhesive that has been pressed with UV light to cure the adhesive; and a second peeling step of releasing pressing of the single-side peeled laminate and peeling the thin film structural body from the single-side peeled laminate to form an optical body formed of the adhesive that has been cured and the thin film structural body on the base plate.

(21) <12> A camera module-equipped device comprising a camera module and a display panel, wherein the display panel includes an adhesive layer stacked on at least part of a surface thereof and a thin film structural body stacked on the adhesive layer, the camera module is arranged such that the camera module and the thin film structural body face each other, and the thin film structural body is a thin film structural body obtained by peeling the first retention film and the second retention film from the laminate according to any one of <1> to <8>.

Advantageous Effect

(22) According to the present disclosure, it is possible to provide a laminate that enables simple peeling of a thin film structural body having a fine irregularity structure at a surface thereof when the thin film structural body is to be bonded to an object serving as an adherend, that has improved environment testing durability, and in which waviness is inhibited. Moreover, according to the present disclosure, it is possible to provide a method of producing this laminate. Furthermore, according to the present disclosure, it is possible to provide a method of forming an optical body having excellent anti-reflection performance on a base plate using this laminate. Also, according to the present disclosure, it is possible to provide a camera module-equipped device that can acquire a captured image in which color irregularities, ghosts, and the like are inhibited.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings:

(2) FIG. 1 is a schematic cross-sectional view illustrating a laminate according to one embodiment of the present disclosure;

(3) FIG. 2 is a top view illustrating a laminate according to one embodiment of the present disclosure;

(4) FIG. 3 is a schematic cross-sectional view illustrating a retention film of a laminate according to one embodiment of the present disclosure;

(5) FIG. 4A is an overview illustrating one step in one example of a method for producing a laminate according to one embodiment of the present disclosure;

(6) FIG. 4B is an overview illustrating one step in one example of a method for producing a laminate according to one embodiment of the present disclosure;

(7) FIG. 4C is an overview illustrating one step in one example of a method for producing a laminate according to one embodiment of the present disclosure;

(8) FIG. 5A is an overview illustrating one step in a method of forming an optical body on a base plate according to one embodiment of the present disclosure;

(9) FIG. 5B is an overview illustrating one step in a method of forming an optical body on a base plate according to one embodiment of the present disclosure;

(10) FIG. 5C is an overview illustrating one step in a method of forming an optical body on a base

plate according to one embodiment of the present disclosure;

(11) FIG. 5D is an overview illustrating one step in a method of forming an optical body on a base plate according to one embodiment of the present disclosure;

(12) FIG. 5E is an overview illustrating one step in a method of forming an optical body on a base plate according to one embodiment of the present disclosure;

(13) FIG. 6 is a schematic overview illustrating a region in proximity to a camera module of a camera module-equipped device according to one embodiment of the present disclosure;

(14) FIG. 7 is a top view illustrating one example of a laminate according to one embodiment of the present disclosure when a punching test is performed using the laminate;

(15) FIG. 8A illustrates the appearance of a laminate according to Comparative Example 2 after testing of detachment upon punching; and

(16) FIG. 8B illustrates the appearance of a laminate according to Example 4 after testing of detachment upon punching.

DETAILED DESCRIPTION

(17) The following provides a detailed description of the present disclosure based on embodiments.

(18) (Laminate)

(19) A presently disclosed laminate includes a thin film structural body and retention films. The thin film structural body has a first retention film stacked on one face of the thin film structural body and has a second retention film stacked on another face of the thin film structural body. The thin film structural body has a first fine irregularity structure at a face that is in contact with the first retention film and has a second fine irregularity structure at a face that is in contact with the second retention film. The first retention film has a third fine irregularity structure at a face that is in contact with the thin film structural body and the second retention film has a fourth fine irregularity structure at a face that is in contact with the thin film structural body. The laminate includes a half cut section in a thickness direction. The second retention film contains a UV-curable resin, and this UV-curable resin has a storage modulus at 25° C. of 1.1 GPa or less.

(20) The presently disclosed laminate serves as an intermediate product used in formation of an optical body on a base plate (described further below).

(21) The following describes a laminate according to one embodiment of the present disclosure (hereinafter, also referred to as the “laminate of the present embodiment”) with reference to FIG. 1, etc.

(22) As illustrated in FIG. 1, the laminate **100** of the present embodiment includes a thin film structural body **110** and two retention films (i.e., a first retention film **120a** and a second retention film **120b**). The first retention film **120a** is stacked on one face of the thin film structural body **110** and the second retention film **120b** is stacked on another face of the thin film structural body **110**. In other words, the thin film structural body **110** is sandwiched between the two retention films. The thin film structural body **110** has a first fine irregularity structure **110-1** at a surface where the first retention film **120a** is stacked and has a second fine irregularity structure **110-2** at a surface where the second retention film **120b** is stacked. Moreover, the first retention film **120a** has a third fine irregularity structure **120a-1** and the second retention film **120b** has a fourth fine irregularity structure **120b-1**.

(23) The laminate **100** of the present embodiment also includes a half cut section **130** in a thickness direction. The “thickness direction” is the direction of stacking of the first retention film **120a**, the thin film structural body **110**, and the second retention film **120b**. The half cut section **130** is a cut line that is formed in order to make it easier for a single-side peeled laminate formed of the thin film structural body **110** and the first retention film **120a** to be peeled from the second retention film **120b**. The half cut section **130** completely severs the first retention film **120a** and the thin film structural body **110** and severs up to partway through the second retention film **120b** as illustrated in FIG. 1.

(24) FIG. 2 is a diagram of the laminate **100** as viewed from above. The half cut section **130** has a

circular shape in a top view as illustrated in FIG. 2. It should be noted, however, that the shape of the half cut section **130** in a top view is not limited to a circular shape. For example, the half cut section **130** may have a polygonal shape such as a quadrangular shape in a top view.

(25) The laminate of the present embodiment satisfies $P1 > P2$ when peeling force at an interface of the first retention film **120a** and the thin film structural body **110** in a 90° peeling test in accordance with JIS Z0237:2009 is taken to be $P1$ (N/25 mm) and peeling force at an interface of the second retention film **120b** and the thin film structural body **110** in a 90° peeling test in accordance with JIS Z0237:2009 is taken to be $P2$ (N/25 mm).

(26) As a result of the laminate **100** of the present embodiment including the half cut section **130** in the thickness direction as previously described, a single-side peeled laminate formed of the thin film structural body **110** and the first retention film **120a** can easily be peeled from the second retention film **120b**, inside of the half cut section **130**, when the thin film structural body **110** is to be bonded to an object serving as an adherend.

(27) Moreover, as a result of the thin film structural body **110** being sandwiched by the two retention films with faces having fine irregularity structures (third fine irregularity structure **120a-1** and fourth fine irregularity structure **120b-1**) in contact therewith in the laminate **100** of the present embodiment as previously described, the thin film structural body **110** can closely adhere to the two retention films without a pressure-sensitive adhesive layer, an adhesive, or the like interposed therebetween. Through the thin film structural body **110** and the two retention films being closely adhered without a pressure-sensitive adhesive layer, an adhesive, or the like interposed therebetween in this manner in the laminate **100** of the present embodiment, the laminate **100** of the present embodiment has high environment testing durability and can inhibit the occurrence of waviness.

(28) Moreover, as a result of there being a difference between one face and the other face of the thin film structural body **110** in terms of peeling force with the retention film stacked thereat as previously described, the laminate **100** of the present embodiment makes it possible to smoothly peel the retention film having low peeling force (second retention film **120b**) from the thin film structural body **110** while maintaining a stacked state of the retention film having high peeling force (first retention film **120a**) and the thin film structural body **110**. Furthermore, damage to a fine irregularity structure is inhibited during peeling of a retention film, and particularly during peeling of the second retention film **120b**. In contrast, when peeling of either of the two retention films is attempted in a case in which peeling force with a retention film is the same at both one face and the other face of the thin film structural body **110**, splitting of the thin film structural body **110** may occur near the center thereof, the majority of a fine irregularity structure may be taken away to the retention film side, and it may not be possible to maintain sufficient quality until the thin film structural body **110** is actually used by a customer.

(29) Note that peeling of the first retention film **120a** can be performed smoothly by, after peeling off the second retention film **120b**, bonding the thin film structural body **110** to a base plate or the like in advance.

(30) In the laminate **100** of the present embodiment, the peeling force $P1$ at the interface of the first retention film **120a** and the thin film structural body **110** and the peeling force $P2$ at the interface of the second retention film **120b** and the thin film structural body **110** are each more than 0 N/25 mm, preferably 0.05 N/25 mm or more, and preferably 0.5 N/25 mm or less. When $P1$ and $P2$ are 0.05 N/25 mm or more, spontaneous peeling of the retention films due to external factors or the like can be inhibited. Moreover, when $P1$ and $P2$ are 0.5 N/25 mm or less, damage to a fine irregularity structure of the thin film structural body **110** can be more sufficiently inhibited during peeling of a retention film.

(31) Note that in a case in which almost the entire surface of the thin film structural body **110** is taken away to the retention film side during measurement of the peeling force at an interface of a retention film and the thin film structural body **110**, it is judged that peeling of the measurement

subject was not possible and that “peel strength measurement value: more than 0 N/25 mm” is not even satisfied.

(32) In the laminate **100** of the present embodiment, the difference (P1–P2) between the peeling force P1 at the interface of the first retention film **120a** and the thin film structural body **110** and the peeling force P2 at the interface of the second retention film **120b** and the thin film structural body is not specifically limited but is preferably 0.03 N/25 mm or more, and is preferably 0.3 N/25 mm or less. When P1–P2 is 0.03 N/25 mm or more, a retention film, and particularly the second retention film **120b**, can be smoothly peeled from the thin film structural body **110** while also more effectively inhibiting damage to the thin film structural body **110** and a fine irregularity structure at the surface thereof. Moreover, when P1–P2 is 0.3 N/25 mm or less, it is possible to more effectively inhibit damage to the thin film structural body **110** and a fine irregularity structure at the surface thereof when the first retention film **120a** is peeled from the thin film structural body.

(33) Note that adjustment of the peeling force at an interface of a retention film and the thin film structural body **110** can be performed as appropriate through operations such as altering a constituent material of the thin film structural body **110**, altering a fine irregularity structure of the thin film structural body **110**, altering the thickness of a part of the thin film structural body **110** that is not a fine irregularity structure (described further below), adding an additive such as a fluorine-containing additive to a constituent material of a base substrate of the retention film (described further below), altering the thickness of the retention film, providing the retention film with an inorganic film, providing the retention film with a release film, and altering a constituent material or the thickness of the inorganic film or the release film. Accordingly, adjustment of P1–P2 can be performed as appropriate through any combination of the operations described above.

(34) <Thin Film Structural Body>

(35) The thin film structural body **110** used in the present embodiment has the first fine irregularity structure **110-1** and the second fine irregularity structure **110-2** at the faces thereof. In other words, fine irregularity patterns (protrusions that protrude in the thickness direction of the laminate and depressions that are depressed in the thickness direction of the laminate) are formed at both faces of the thin film structural body **110**. The protrusions and depressions may be arranged periodically (for example, a staggered or rectangular grid), or may be arranged randomly. Moreover, the shape of the protrusions and the depressions is not specifically limited and may be a bullet shape, a pyramidal shape, a columnar shape, a needle shape, or the like. Note that the shape of a depression is the shape formed by an inner wall of the depression.

(36) The thin film structural body **110** can be produced using a UV (ultraviolet) curable resin, for example. The UV-curable resin may be a UV-curable acrylic-based resin, a UV-curable epoxy-based resin, or the like, for example, without any specific limitations.

(37) The average period (pitch) of irregularity patterns constituting the first fine irregularity structure **110-1** and the second fine irregularity structure **110-2** is preferably equal to or less than a wavelength of visible light (for example, 830 nm or less), more preferably 350 nm or less, and even more preferably 280 nm or less, and is more preferably 100 nm or more, and even more preferably 150 nm or more. By setting the pitch of an irregularity pattern at a surface of the thin film structural body **110** as equal to or less than a wavelength of visible light (i.e., by providing a structure referred to as a “moth-eye structure” at the surface of the thin film structural body **110**), further improvement of anti-reflection performance can be achieved.

(38) The depth of depressions (height of protrusions) of the irregularity patterns constituting the first fine irregularity structure **110-1** and the second fine irregularity structure **110-2** is not specifically limited but is preferably 150 nm or more, and more preferably 190 nm or more, and is preferably 300 nm or less, and more preferably 230 nm or less.

(39) Note that the arrangement of depressions and protrusions, the average period of the irregularity pattern, the depth of depressions, and so forth may be the same or different for the first

fine irregularity structure **110-1** and the second fine irregularity structure **110-2** of the thin film structural body **110**.

(40) The thickness of the thin film structural body **110** used in the present embodiment is preferably 10 μm or less, more preferably 5 μm or less, and even more preferably 1 μm or less, and is preferably 0.1 μm or more. Such a thin film structural body **110** can suitably be used in an application where both improvement of anti-reflection performance and reduction of thickness are required, such as in a notebook PC, tablet PC, smartphone, mobile phone, or the like that is equipped with an image sensor.

(41) Note that the thickness of the thin film structural body **110** is the distance in the stacking direction or film thickness direction between the apex of a highest protrusion formed at one face and an apex of a highest protrusion formed at the other face as indicated by T.sub.1 in FIG. 1.

(42) The thickness of a part that is not a fine irregularity structure in the thin film structural body **110** used in the present embodiment is preferably 200 nm or less. When a laminate **100** including such a thin film structural body **110** is used in formation of an optical body by a subsequently described method of forming an optical body on a base plate according to one embodiment of the present disclosure, for example, the thin film structural body **110** can be more reliably peeled from a single-side peeled laminate after being bonded onto a base plate through an adhesive, and an optical body can be formed with high precision. On the other hand, the thickness of the part of the thin film structural body **110** that is not a fine irregularity structure can be set as 0.01 μm or more from a viewpoint of feasibility.

(43) Note that the thickness of the part of the thin film structural body **110** that is not a fine irregularity structure is the distance in the stacking direction or film thickness direction between an apex of a deepest depression formed at one face and an apex of a deepest depression formed at the other face as indicated by T.sub.2 in FIG. 1.

(44) The thin film structural body **110** can be produced by, for example, preparing a base substrate and two fine irregularity layers as separate members and then forming the fine irregularity layers at both faces of the base substrate. However, it is preferable that the thin film structural body **110** is formed of a single member from a viewpoint of avoiding deterioration of optical properties. A laminate **100** that includes a thin film structural body **110** formed of a single member in this manner can be produced by the subsequently described method of producing a laminate **100**, for example.

(45) <Retention Films>

(46) In the laminate **100** of the present embodiment, the two retention films (i.e., the first retention film **120a** and the second retention film **120b**) sandwich the thin film structural body **110**. These two retention films are provided for protection of the thin film structural body **110**, improvement of handleability, and so forth.

(47) As illustrated in FIG. 1, the first retention film **120a** has a third fine irregularity structure **120a-1** at a face that is in contact with the thin film structural body **110** and the second retention film **120b** has a fourth fine irregularity structure **120b-1** at a face that is in contact with the thin film structural body **110**. In other words, a fine irregularity pattern (protrusions that protrude in the thickness direction of the laminate **100** and depressions that are depressed in the thickness direction of the laminate **100**) is formed at a specific face of the first retention film **120a** and a specific face of the second retention film **120b**. This makes it simple to form fine irregularity structures at both faces of the thin film structural body **110**.

(48) Moreover, as illustrated in FIG. 1, the third fine irregularity structure **120a-1** of the first retention film **120a** is preferably an inverted structure of the first fine irregularity structure **110-1** of the thin film structural body **110**, and the fourth fine irregularity structure **120b-1** of the second retention film **120b** is preferably an inverted structure of the second fine irregularity structure **110-2** of the thin film structural body **110**. This makes it possible for the thin film structural body **110** and the retention films **120a** and **120b** to be mechanically joined through the respective fine irregularity

structures thereof and to strengthen protection of the fine irregularity structures formed at both faces of the thin film structural body **110**, while also more effectively inhibiting damage to the thin film structural body **110** during peeling of a retention film. From the same viewpoints, it is more preferable that the fine irregularity structure of the first retention film **120a** and the fine irregularity structure of the second retention film **120b** interlock, without gaps, with the fine irregularity structures of the thin film structural body **110**. Moreover, the thin film structural body **110** can closely adhere to the retention films **120a** and **120b** through frictional force due to mechanical joining through the fine irregularity structures of both the thin film structural body **110** and the retention films **120a** and **120b**. This effect of close adherence through frictional force due to interlocking of fine irregularity structures without gaps is referred to as an “anchor effect”. Through this anchor effect, it is possible to maintain a closely adhered state of the thin film structural body **110** with the retention films **120a** and **120b** even without a pressure-sensitive adhesive, an adhesive, or the like interposed therebetween.

(49) The average period (pitch) of the irregularity patterns constituting the third fine irregularity structure **120a-1** and the fourth fine irregularity structure **120b-1** is preferably equal to or less than a wavelength of visible light (for example, 830 nm or less), more preferably 350 nm or less, and even more preferably 280 nm or less, and is more preferably 100 nm or more, and even more preferably 150 nm or more in the same way as for the first fine irregularity structure **110-1** and the second fine irregularity structure **110-2**.

(50) The retention films **120a** and **120b** can each be produced from a base substrate, for example. Moreover, the retention films **120a** and **120b** that each have a fine irregularity structure at a surface thereof can be produced as illustrated in FIG. 3, for example, by forming a fine irregularity layer **122** on a base substrate **121**.

(51) Although no specific limitations are placed on the material forming the base substrate **121**, the material is preferably a material that is transparent and does not easily rupture, and may be PET (polyethylene terephthalate), TAC (triacetyl cellulose), or the like.

(52) Formation of the fine irregularity layer **122** on the base substrate **121** can be achieved by, for example, implementing a method including a step of applying an uncured UV-curable resin onto one face of the base substrate **121**, a step of bringing a roll on which a corresponding irregularity pattern has been formed into close contact with the applied UV-curable resin and transferring an irregularity pattern to the UV-curable resin, a step of irradiating the applied UV-curable resin with UV light to cure the UV-curable resin, and a step of peeling the UV-curable resin that has been cured from the roll. Note that the UV-curable resin may be a UV-curable acrylic-based resin, a UV-curable epoxy-based resin, or the like, for example, without any specific limitations. Moreover, various additives such as curing initiators may be added to the UV-curable resin as necessary.

(53) <Production Method of Laminate>

(54) The method by which the laminate **100** of the present embodiment is produced is not specifically limited and can be selected as appropriate depending on the objective. The following describes a specific example of a method for producing the laminate **100** with reference to FIGS. 4A to 4C.

(55) The one example of a method includes a step of sandwiching a UV-curable resin between two retention films each having a fine irregularity structure at a surface thereof and performing pressure bonding thereof (sandwiching and pressure bonding step), a step of irradiating the UV-curable resin that has been sandwiched with UV light to cure the UV-curable resin (curing step), and a step of forming a half cut section **130** (half cut section forming step).

(56) —Sandwiching and Pressure Bonding Step—

(57) First, two retention films each having a fine irregularity structure at a surface thereof (first retention film **120a** having third fine irregularity structure **120a-1** at surface and second retention film **120b** having fourth fine irregularity structure **120b-1** at surface) are prepared. The first retention film **120a** and the second retention film **120b** are as previously described. Next, a UV-

curable resin **151** is sandwiched between the above-described first retention film **120a** and second retention film **120b** such that the fine irregularity structures of the first retention film **120a** and the second retention film **120b** face each other as illustrated in FIG. 4A. Note that the UV-curable resin may be a UV-curable acrylic-based resin, a UV-curable epoxy-based resin, or the like, for example, without any specific limitations. Moreover, various additives such as curing initiators may be added to the UV-curable resin **151** as necessary. Furthermore, a monomer such as an ethylene oxide-based (EO-based) acrylic monomer, a propylene oxide-based (PO-based) acrylic monomer, or a fluorene-based monomer may be added to the UV-curable resin **151** from a viewpoint of increasing peelability and shape retainability.

(58) The viscosity of the UV-curable resin **151** is preferably 30 cps or less. When the viscosity of the UV-curable resin **151** is 30 cps or less, it is possible to reduce the thickness of a part that is not a fine irregularity structure to 200 nm or less more easily in formation of the thin film structural body **110**.

(59) As illustrated in FIG. 4A, the sandwiched body is pressure bonded in a sandwiching direction through a pressure bonding device such as a roll laminator **160**. By adjusting the pressure during pressure bonding in the sandwiching and pressure bonding step, it is possible to adjust the thickness of the obtained thin film structural body **110** (T.sub.1 in FIG. 1) and the thickness of the part of the thin film structural body **110** that is not a fine irregularity structure (T.sub.2 in FIG. 1). The thickness of the obtained thin film structural body **110** (T.sub.1 in FIG. 1) and the thickness of the part of the thin film structural body **110** that is not a fine irregularity structure (T.sub.2 in FIG. 1) can also be adjusted by adjusting the feed rate by the roll laminator.

(60) It should be noted that although the second retention film **120b** is positioned at a lower side and the first retention film **120a** is positioned at an upper side relative to the roll laminator **160** in FIG. 4A, the positional relationship thereof is not specifically limited.

(61) —Curing Step—

(62) In the curing step, the UV-curable resin **151** that has been sandwiched is irradiated with UV light to cure the UV-curable resin **151** as illustrated in FIG. 4B. By curing the UV-curable resin **151**, a thin film structural body **110** having fine irregularity structures at both faces, such as illustrated in FIG. 1, is formed as a single member, and a laminate **100** is obtained. Note that the curing step may be performed at the same timing as the sandwiching and pressure bonding step.

(63) In the thin film structural body **110** that is obtained in this manner, a fine irregularity structure (first fine irregularity structure **110-1**) that is interlocked, without gaps, with the third fine irregularity structure **120a-1** of the first retention film **120a** is formed at one face of the thin film structural body **110**, and a fine irregularity structure (second fine irregularity structure **110-2**) that is interlocked, without gaps, with the fourth fine irregularity structure **120b-1** of the second retention film **120b** is formed at the other face of the thin film structural body **110**.

(64) —Half Cut Section Forming Step—

(65) In the half cut section forming step, a half cut section **130** is formed by, in a state in which the laminate **100** is pressed in a direction of arrow A by an air cylinder **132**, punching the periphery of the pressed part using a blade **131** as illustrated in FIG. 4C. The air cylinder **132** keeps the laminate **100** in a pressed state until the blade **131** has been withdrawn from the laminate **100**. By pressing the laminate **100** using the air cylinder **132** in this manner, it is possible to prevent the first retention film **120a** and the thin film structural body **110** being taken away by the blade **131** and the thin film structural body **110** being peeled from the second retention film **120b**, inside of the half cut section **130**, when the blade **131** is withdrawn.

(66) When a half cut section is formed by laser cutting, for example, a lot of debris is formed, but since the half cut section **130** is formed through punching using the blade **131** in the half cut section forming step according to the present disclosure, the formation of debris during formation of the half cut section **130** can be inhibited.

(67) Moreover, uprising of approximately 10% of film thickness at a severed part arises when a

half cut section is formed by laser cutting, but by making the blade **131** a single-edged blade in the half cut section forming step according to the present disclosure, uprising of a severed part inside of the half cut section **130** can be reduced.

(68) (Method of Forming Optical Body on Base Plate)

(69) A presently disclosed method of forming an optical body on a base plate includes peeling the first retention film **120a** and the second retention film **120b** from the presently disclosed laminate **100** set forth above and stacking the thin film structural body **110** on a base plate via an adhesive. Through this method, it is possible for a thin film structural body **110** having fine irregularity structures at both faces to be formed, without damage, on a base plate as an optical body having excellent anti-reflection performance. In particular, through this method, it is possible to significantly inhibit damage to the fine irregularity structures during formation of the optical body even in a case in which the thin film structural body **110** is extremely thin (for example, 5 μm or less).

(70) The following describes a method of forming an optical body on a base plate according to one embodiment of the present disclosure (hereinafter, also referred to as the “formation method of the present embodiment”) with reference to FIGS. 5A to 5E.

(71) FIGS. 5A to 5E are overviews illustrating the formation method of the present embodiment. The formation method of the present embodiment includes a first peeling step, an application step, a pressing step, a curing step, and a second peeling step.

(72) In the first peeling step, a single-side peeled laminate formed of the thin film structural body **110** and the first retention film **120a** is peeled from the second retention film **120b** illustrated in FIG. 1, inside of the half cut section **130**, and is placed in a state (single-side peeled laminate **100'**) illustrated in FIG. 5A. As a result of P2 (peeling force at interface of second retention film **120b** and thin film structural body **110**) being smaller than P1 (peeling force at interface of first retention film **120a** and thin film structural body **110**), the single-side peeled laminate can be smoothly peeled from the second retention film **120b** while also maintaining a stacked state of the first retention film **120a** and the thin film structural body **110**, and inhibiting damage to the second fine irregularity structure **110-2**.

(73) After the first peeling step, an adhesive **171** is applied onto a base plate **170** in the application step as illustrated in FIG. 5B. The adhesive **171** is not specifically limited and may, for example, be a composition containing a UV-curable resin such as a UV-curable acrylic-based resin or a UV-curable epoxy-based resin. The material forming the base plate **170** is not specifically limited and can be selected as appropriate depending on the objective of forming the optical body. For example, the material may be glass, glass that is surface coated with any organic material (for example, an epoxy acrylate copolymer), polymethyl methacrylate (PMMA), a cycloolefin copolymer (COC), a cycloolefin polymer (COP), or the like.

(74) After the application step, the single-side peeled laminate **100'** is pressed against the adhesive **171** that has been applied onto the base plate **170** such that the face where the second retention film **120b** has been peeled off faces toward the base plate **170** in the pressing step as illustrated in FIG. 5C. The pressed adhesive **171** spreads out between the base plate **170** and the thin film structural body **110**.

(75) In the curing step illustrated in FIG. 5D, the pressed adhesive **171** is irradiated with UV light to cure the adhesive **171** while in a state in which pressing is maintained. The cured adhesive **171** strongly adheres to the base plate **170** and the thin film structural body **110**.

(76) In the second peeling step illustrated in FIG. 5E, pressing of the single-side peeled laminate **100'** is released in order to peel the first retention film **120a** from the thin film structural body **110**. Through the method set forth above, it is possible to form a thin film structural body **110** (or optical body **200**) in a partial manner and with high precision with respect to a target region of a base plate surface. In particular, the method set forth above is advantageous in terms that a thin film structural body **110** (or optical body **200**) can be formed in a partial manner and with high precision even

with respect to a base plate to which adhesion is difficult (for example, a base plate formed of a polymer such as a cycloolefin copolymer (COC), a cycloolefin polymer (COP), or an epoxy acrylate copolymer).

(77) In the optical body **200** that is formed on the base plate **170**, the cured adhesive **171'** can also enter depressions in a face at the base plate **170** side of the thin film structural body **110** as illustrated in FIG. 5E. In other words, the cured adhesive **171'** can have a fine irregularity structure at a surface at the thin film structural body **110** side thereof. Such an optical body **200** has excellent anti-reflection performance and can, for example, have an average reflectance of 1% or less in a wavelength range of 400 nm to 750 nm.

(78) (Camera Module-Equipped Device)

(79) A presently disclosed camera module-equipped device includes a camera module and a display panel, wherein the display panel includes an adhesive layer stacked on at least part of a surface thereof and a thin film structural body stacked on the adhesive layer. The thin film structural body has fine irregularity structures at both faces thereof. In this camera module-equipped device, the camera module is arranged such that the camera module and the thin film structural body face each other. Through this camera module-equipped device, an image sensor of the camera module can capture a still image or a video via the thin film structural body having fine irregularity structures at both faces, and thus reflection of light can be suppressed, and color irregularities, ghosts, and the like in a captured image that is acquired can be inhibited.

(80) The camera module-equipped device may, more specifically, be a notebook PC, a tablet PC, a smartphone, a mobile phone, or the like.

(81) The following describes a camera module-equipped device according to one embodiment of the present disclosure (hereinafter, also referred to as the “device according to the present embodiment”) with reference to FIG. 6.

(82) FIG. 6 is a schematic overview illustrating a region in proximity to a camera module of the camera module-equipped device of the present embodiment. As illustrated in FIG. 6, the camera module-equipped device **300** of the present embodiment includes a camera module **310** and a display panel **311**. A light shielding region **312** and a transparent region (non-light shielding region) **313** are formed at one surface of the display panel **311**. In the transparent region **313** of the display panel **311**, an adhesive layer **314** is stacked and, in addition thereto, a thin film structural body **315** is stacked on the adhesive layer **314**.

(83) The display panel **311** is preferably transparent in order that it can be used as a liquid-crystal display, a touch panel, or the like. For example, the display panel **311** may be formed of glass, glass that is surface coated with any organic material, polymethyl methacrylate (PMMA), or the like. The thin film structural body **315** is as previously described for the thin film structural body **110** included in the presently disclosed laminate **100** set forth above.

(84) The adhesive layer **314** and the thin film structural body **315** can be formed on the display panel **311**, serving as a base plate, by the presently disclosed method of forming an optical body on a base plate set forth above, and using the presently disclosed laminate **100** set forth above. In such a situation, the adhesive layer **314** and the thin film structural body **315** respectively correspond to the cured adhesive **171'** and the thin film structural body **110** described for the formation method of the present embodiment.

(85) The camera module **310** is arranged such that the camera module **310** and the thin film structural body **315** face each other as illustrated in FIG. 6.

(86) No specific limitations are placed on detailed conditions of the device of the present embodiment, such as the specific configuration of the camera module **310**, the distance between the camera module **310** and the thin film structural body **315**, and so forth.

Examples

(87) The following provides a more specific description of the present disclosure through examples and comparative examples. However, the present disclosure is not limited to the following

examples.

(88) Note that testing of detachment upon punching and testing of detachment after vibration were performed by the following procedures for laminates produced in the examples and comparative examples. In addition, half cut processing was evaluated based on the results of testing of detachment upon punching and testing of detachment after vibration.

(89) (Testing of Detachment Upon Punching)

(90) Half cut sections **130** were formed at 20 locations in 5 rows by 4 columns in a laminate **100** as illustrated in FIG. 7. When each of these half cut sections **130** was formed, it was confirmed whether or not a single-side peeled laminate inside of the half cut section **130** became detached.

(91) In testing of detachment upon punching, the outer dimensions of the laminate **100** were set as 50 mm in length by 40 mm in width. Moreover, the interval between centers of adjacent half cut sections **130** was set as 9 mm.

(92) Conditions of a pressing machine, air cylinder **132**, and blade **131** in formation of the half cut sections **130** were set as indicated below. Pressing machine pressure: 35 kN Air cylinder pressure (per one cylinder inside circle): 19.7 N Blade: Thompson blade, single-edged blade specification Pressing machine model no.: MKP-150H produced by Mikado Technos Co., Ltd.

(Testing of Detachment after Vibration)

(93) Testing of detachment after vibration was performed for laminates **100** for which detachment of a single-side peeled laminate inside of a half cut section **130** did not occur in testing of detachment upon punching. In the testing of detachment after vibration, the laminate **100** was subjected to vibration anticipated to occur during transportation thereof, and it was confirmed whether or not a single-side peeled laminate inside of the half cut section **130** became detached.

(94) The testing of detachment after vibration was performed by subjecting a laminate **100** packaged in corrugated cardboard or the like to vibration under the following conditions. Power spectral density: 5 Hz to 50 Hz, 1.44 [s²/m³] Overall effective value: 0.83 G Vibration time: 20 minutes/axis

(Evaluation of Half Cut Processing)

(95) Half cut processing was evaluated by the following standard based on the results of testing of detachment upon punching and testing of detachment after vibration.

(96) A: No detachment in both testing of detachment upon punching and testing of detachment after vibration

(97) B: No detachment in testing of detachment upon punching but detachment in testing of detachment after vibration

(98) C: Detachment in testing of detachment upon punching

(99) As indicated above, a laminate **100** for which detachment did not occur in testing of detachment upon punching but did occur in testing of detachment after vibration was given a B evaluation rather than a C evaluation because such a laminate **100** can be transported without detachment by transporting the laminate **100** under transportation conditions with little vibration.

(100) (Material of Retention Film)

(101) The following material was used as the base substrate of a retention film in a laminate for which testing of detachment upon punching and testing of detachment after vibration were performed. Note that the following material has four thicknesses (50 μ m, 75 μ m, 100 μ m, and 125 μ m), and thus a film of one or other of these thicknesses was used. Base substrate: COSMOSHINE A4300 PET film produced by Toyobo Co., Ltd.

(102) Moreover, one or other of the three types of resins shown below was used as a fine irregularity layer of the retention film (i.e., a UV-curable resin contained in the retention film). Note that the following resins 1 to 3 are resins that differ in terms of storage modulus at 25° C. Resin 1 has a storage modulus at 25° C. of 4.2 GPa, resin 2 has a storage modulus at 25° C. of 1.1 GPa, and resin 3 has a storage modulus at 25° C. of 510 MPa. Resin 1 (storage modulus at 25° C.: 4.2 GPa) UVX6366 produced by Toagosei Co., Ltd.: 59% M240 produced by Toagosei Co., Ltd.:

39% Irgacure 184 produced by BASF Japan: 2% Resin 2 (storage modulus at 25° C.: 1.1 GPa) UVX6366 produced by Toagosei Co., Ltd.: 59% UN-6200 produced by Negami Chemical Industrial Co., Ltd.: 34% Viscoat #150 (THFA) produced by Osaka Organic Chemical Industry Ltd.: 5% Irgacure 184 produced by BASF Japan: 2% Resin 3 (storage modulus at 25° C.: 510 MPa) UVX6366 produced by Toagosei Co., Ltd.: 49% Produced by Kyoeisha Chemical Co., Ltd.: 49% Irgacure 184 produced by BASF Japan: 2%

<Dependence on Thickness of Base Substrate of First Retention Film>

(103) Testing of detachment upon punching and testing of detachment after vibration were performed with the thickness of a base substrate **121** of a first retention film **120a** set as the four values indicated below. Example 1: 50 μm Example 2: 75 μm Example 3: 100 μm Comparative Example 1: 125 μm

(104) In this testing, the thickness of a base substrate **121** of a second retention film **120b**, the storage modulus of a fine irregularity layer **122** of the second retention film **120b**, and the storage modulus of a fine irregularity layer **122** of the first retention film **120a** were set as follows as conditions that were common for each example. Thickness of base substrate of second retention film: 125 μm Storage modulus of fine irregularity layer **122** of second retention film: 1.1 GPa (resin 2 used) Storage modulus of fine irregularity layer **122** of first retention film: 4.2 GPa (resin 1 used)

(105) Results for the dependence on thickness of the base substrate of the first retention film performed for Examples 1 to 3 and Comparative Example 1, described above, are shown below in Table 1.

(106) TABLE-US-00001

	TABLE 1	Comparative Example 1	Example 2	Example 3	Example 1
Thickness of base	125 μm	125 μm	125 μm	125 μm	substrate of second retention film
Thickness of base	50 μm	75 μm	100 μm	125 μm	substrate of first retention film
Storage modulus of 1.1 GPa	1.1 GPa	1.1 GPa	1.1 GPa	second retention film (resin 2)	(resin 2)
Storage modulus of 4.2 GPa	4.2 GPa	4.2 GPa	4.2 GPa	4.2 GPa	first retention film (resin 1)
Detachment upon OK	OK	OK	OK	OK	punching
Detachment after OK	OK	OK	OK	Poor	vibration
Evaluation of half cut	A	A	A	B	processing

(107) In “Detachment upon punching” or “Detachment after vibration” of Table 1, “OK” indicates that detachment did not occur, whereas “Poor” indicates that detachment did occur.

(108) As shown in Table 1, detachment did not occur in testing of detachment upon punching or in testing of detachment after vibration in Examples 1 to 3. Accordingly, a good result (A evaluation) was given in evaluation of half cut processing. This is thought to be due to the base substrate **121** of the first retention film **120a** having sufficient flexibility as a result of being thin, and due to deformation of the first retention film **120a** during punching being small. It is also thought that as a result of deformation of the first retention film **120a** being small, close adherence through an anchor effect between the thin film structural body **110** and the second retention film **120b** was maintained, and, consequently, detachment did not occur even in testing of detachment after vibration.

(109) On the other hand, although detachment did not occur in testing of detachment upon punching in Comparative Example 1, detachment did occur in testing of detachment after vibration. Accordingly, a B evaluation was given in evaluation of half cut processing. This is thought to be due to the base substrate **121** of the first retention film **120a** having insufficient flexibility as a result of being thick, and due to deformation of the first retention film **120a** during punching being large. It is also thought that upon punching, the thin film structural body **110** was peeled from the second retention film **120b** to a degree that did not cause detachment, and the anchor effect between the thin film structural body **110** and the second retention film **120b** was lost due to deformation of the first retention film **120a** being large, and thus detachment did occur in testing of detachment after vibration.

(110) <Dependence on Storage Modulus of Second Retention Film>

(111) Testing of detachment upon punching and testing of detachment after vibration were performed with the storage modulus of a fine irregularity layer **122** of a second retention film **120b** set as the following three values. Example 4: 510 MPa (resin 3 used) Example 1: 1.1 GPa (resin 2 used) Comparative Example 2: 4.2 GPa (resin 1 used)

(112) In this testing, the thickness of a base substrate **121** of the second retention film **120b**, the thickness of a base substrate **121** of a first retention film **120a**, and the storage modulus of a fine irregularity layer **122** of the first retention film **120a** were set as follows as conditions that were common for each example. Thickness of base substrate of second retention film: 125 μm Thickness of base substrate of first retention film: 50 μm Storage modulus of fine irregularity layer of first retention film: 4.2 GPa (resin 1 used)

(113) Results for the dependence on the storage modulus of the fine irregularity layer of the second retention film performed for Example 4, Example 1, and Comparative Example 2, described above, are shown below in Table 2.

(114) TABLE-US-00002

	Comparative Example 4	Example 1	Example 2
Thickness of base substrate of second retention film	125 μm	125 μm	125 μm
Thickness of base substrate of first retention film	50 μm	50 μm	50 μm
Storage modulus of second retention film (resin 3)	510 MPa	1.1 GPa	4.2 GPa
Storage modulus of first retention film (resin 1)	4.2 GPa	4.2 GPa	4.2 GPa
Detachment upon punching	OK	OK	Poor
Detachment after vibration	OK	OK	N/A
Evaluation of half cut processing	A	A	C

(115) FIG. **8A** illustrates the appearance of a laminate **100** according to Comparative Example 2 after testing of detachment upon punching. In the laminate **100** according to Comparative Example 2, detachment occurred for all of the half cut sections **130** at the 20 locations in 5 rows by 4 columns as illustrated in FIG. **8A**. FIG. **8B** illustrates the appearance of a laminate **100** according to Example 4 after testing of detachment upon punching. In the laminate **100** according to Example 4, detachment did not occur for any of the half cut sections **130** at the 20 locations in 5 rows by 4 columns as illustrated in FIG. **8B**.

(116) In “Detachment upon punching” or “Detachment after vibration” of Table 2, “OK” indicates that detachment did not occur, whereas “Poor” indicates that detachment did occur. Moreover, “N/A” for “Detachment after vibration” indicates that detachment occurred in testing of detachment upon punching, and thus testing of detachment after vibration was not subsequently performed.

(117) As shown in Table 2, detachment did not occur in testing of detachment upon punching or testing of detachment after vibration in Example 4 and Example 1. Accordingly, a good result (A evaluation) was given in evaluation of half cut processing. On the other hand, detachment occurred in testing of detachment upon punching in Comparative Example 2. Accordingly, a C evaluation was given in evaluation of half cut processing. Based on the results shown in Table 2, it is thought that a good evaluation of half cut processing is achieved when the storage modulus of the fine irregularity layer **122** of the second retention film **120b** is 1.1 GPa or less.

(118) Through evaluation of “Dependence on thickness of base substrate of first retention film” and “Dependence on storage modulus of second retention film” described above, it was possible to discover suitable conditions for forming the half cut section **130** in the laminate **100**. Specifically, the thickness of the base substrate **121** of the first retention film **120a** is preferably 100 μm or less. Moreover, the storage modulus at 25° C. of the fine irregularity layer **122** of the second retention film **120b** is preferably 1.1 GPa or less.

INDUSTRIAL APPLICABILITY

(119) According to the present disclosure, it is possible to provide a laminate that enables simple peeling of a thin film structural body having a fine irregularity structure at a surface thereof when the thin film structural body is to be bonded to an object serving as an adherend, that has improved environment testing durability, and in which waviness is inhibited. Moreover, according to the present disclosure, it is possible to provide a method of producing this laminate. Furthermore,

according to the present disclosure, it is possible to provide a method of forming an optical body having excellent anti-reflection performance on a base plate using this laminate. Also, according to the present disclosure, it is possible to provide a camera module-equipped device that can acquire a captured image in which color irregularities, ghosts, and the like are inhibited.

REFERENCE SIGNS LIST

(120) **100** laminate **100'** single-side peeled laminate **110** thin film structural body **110-1** first fine irregularity structure **110-2** second fine irregularity structure **120a** first retention film **120a-1** third fine irregularity structure **120b** second retention film **120b-1** fourth fine irregularity structure **121** base substrate **122** fine irregularity layer **130** half cut section **131** blade **132** air cylinder **151** UV-curable resin **160** roll laminator **170** base plate **171** adhesive **171'** cured adhesive **200** optical body **300** camera module-equipped device **310** camera module **311** display panel **312** light shielding region **313** transparent region **314** adhesive layer **315** thin film structural body

Claims

1. A laminate comprising a film structural body and retention films, wherein a first retention film is stacked on one face of the film structural body and a second retention film is stacked on another face of the film structural body, the film structural body has a first irregularity structure at a face that is in contact with the first retention film and has a second irregularity structure at a face that is in contact with the second retention film, the first retention film has a third irregularity structure at a face that is in contact with the film structural body, the second retention film has a fourth irregularity structure at a face that is in contact with the film structural body, the laminate includes a half cut section in a thickness direction, the second retention film contains a UV-curable resin, and the UV-curable resin has a storage modulus at 25° C. of 1.1 GPa or less, the first retention film and the second retention film are peelable from the film structural body, $P1 > P2$ is satisfied when peeling force at an interface of the first retention film and the film structural body in a 90° peeling test in accordance with JIS Z0237:2009 is taken to be $P1$ (N/25 mm) and peeling force at an interface of the second retention film and the film structural body in a 90° peeling test in accordance with JIS Z0237:2009 is taken to be $P2$ (N/25 mm), and $P1 - P2$ is 0.03 (N/25 mm) or more and is 0.3 (N/25 mm) or less.
2. The laminate according to claim 1, wherein a base substrate of the first retention film has a thickness of 100 μ m or less.
3. The laminate according to claim 1, wherein the half cut section has a circular shape or a polygonal shape in a top view of the laminate.
4. The laminate according to claim 1, wherein the film structural body has a thickness of 5 μ m or less.
5. The laminate according to claim 1, wherein the first retention film contains a UV-curable resin.
6. The laminate according to claim 1, wherein the third irregularity structure is an inverted structure of the first irregularity structure and the fourth irregularity structure is an inverted structure of the second irregularity structure.
7. The laminate according to claim 1, wherein the first irregularity structure, the second irregularity structure, the third irregularity structure, and the fourth irregularity structure are each formed of an irregularity pattern having a pitch equal to or less than a wavelength of visible light.
8. A method of producing a laminate comprising a step of, in formation of the half cut section in the laminate according to claim 1, forming the half cut section with the laminate in a pressed state.
9. A method of forming an optical body on a base plate comprising peeling the first retention film and the second retention film from the laminate according to claim 1 and stacking the film structural body on a base plate via an adhesive.
10. The method of forming an optical body on a base plate according to claim 9 comprising: a first peeling step of peeling a single-side peeled laminate formed of the film structural body and the first

retention film from the second retention film, inside of the half cut section, to obtain the single-side peeled laminate; an application step of applying the adhesive onto the base plate; a pressing step of pressing the single-side peeled laminate against the adhesive that has been applied such that a face where the second retention film has been peeled off faces toward the base plate; a curing step of irradiating the adhesive that has been pressed with UV light to cure the adhesive; and a second peeling step of releasing pressing of the single-side peeled laminate and peeling the film structural body from the single-side peeled laminate to form an optical body formed of the adhesive that has been cured and the film structural body on the base plate.

11. A camera module-equipped device comprising a camera module and a display panel, wherein the display panel includes an adhesive layer stacked on at least part of a surface thereof and a film structural body stacked on the adhesive layer, the camera module is arranged such that the camera module and the film structural body face each other, and the film structural body is a film structural body obtained by peeling the first retention film and the second retention film from the laminate according to claim 1.
